

## 0.1 Heterogeneous Effects by Pre-Pandemic Religiosity

To study whether the effect of COVID-19 lockdowns on religious search behavior varied with pre-existing levels of religiosity, I split states into high- and low-religiosity groups based on the median of the *percentage of residents classified as “very religious”* in the 2017 Gallup U.S. Daily survey—the most recent year for which Gallup published state-level religiosity estimates from its large-scale tracking program.<sup>1</sup> Before presenting results, I address a potential methodological concern—the *ceiling effect*—and explain why the design of this paper effectively rules it out.

### 0.1.1 Why a Ceiling Effect Might Be a Concern

A ceiling effect arises in a difference-in-differences setting when units assigned to one group are already operating near the maximum of the outcome scale prior to treatment, leaving insufficient room to record an increase even if the underlying behavioral response is genuinely large. In the present context, the concern would run as follows. If residents of high-religiosity states already engage in religious search at near-maximum intensity before COVID-19—reflected in RSI values close to 100 during the pre-lockdown window of 2019—then the normalized index has little upward headroom. A large lockdown-induced impulse toward religious activity might therefore manifest only as a small measured increase, *not* because the behavioral response is genuinely smaller in high-religiosity states, but simply because the measurement instrument is constrained from above. Concretely, suppose that in a high-religiosity state the average RSI in the pre-treatment 2019 window is 85 out of 100. Even if lockdowns trigger an underlying response of 20 index points, the observed increase can be at most 15 points. A low-religiosity state starting at RSI 40, by contrast, faces no such constraint and can fully register a 20-point increase. If high-religiosity states systematically have higher baselines, then the estimated heterogeneity coefficient  $\hat{\phi}$  in

$$y_{it} = \gamma(D_{it} \times \mathbf{1}_{2020}) + \delta D_{it} + \phi(D_{it} \times \mathbf{1}_{2020} \times \text{High}_i) + \psi(D_{it} \times \text{High}_i) + \theta(\text{High}_i \times \mathbf{1}_{2020}) + \mu_i + \lambda_t + \varepsilon_{it} \quad (1)$$

would be negative and significant even in the absence of any true differential behavioral response—a spurious finding driven entirely by the bounded support of the outcome variable.

### 0.1.2 Why the Ceiling Effect Is Off the Table in This Design

The concern described above does not apply here, for a reason rooted in the normalization procedure introduced in Section 3 (following ?). Recall that for each state  $i$ , the 2019 and 2020 daily RSI series are made jointly comparable by rescaling both to the same denominator—the maximum RSI across *both* years combined:

$$\tilde{d}_t = \frac{c_t}{\max\{(c_t)_{t \in \mathcal{T}}, (c'_t)_{t \in \mathcal{T}'}\}}, \quad \tilde{d}'_t = \frac{c'_t}{\max\{(c_t)_{t \in \mathcal{T}}, (c'_t)_{t \in \mathcal{T}'}\}}, \quad (2)$$

where  $c_t$  and  $c'_t$  are the weekly-rescaled series for 2020 and 2019, respectively. This normalization has a critical implication: the *ceiling is set jointly by both years within each state*. If lockdowns in 2020 generate a large surge in religious search in state  $i$ , the maximum of the combined series in (2) will itself be driven by that 2020 peak, automatically expanding the scale to accommodate it. The 2019 baseline will then be rescaled downward relative to this 2020-driven ceiling. The result is that no post-treatment observation in 2020 is structurally constrained from above by pre-treatment behavior in 2019.

More intuitively: for the ceiling effect to distort the DiD estimate, the 2019 pre-treatment RSI would need to be close to 100 under the *joint* normalization. But by construction, a value of 100 in the joint series is achieved by the single highest observation across both years. If that highest observation occurs in 2020 (which is precisely the case when lockdowns drive a large response), then 2019 values are well below 100 and ample headroom exists for the DiD to detect the increase. The normalization procedure therefore *endogenously* prevents the ceiling from binding in the direction that would cause concern.

<sup>1</sup>Gallup discontinued the U.S. Daily tracking survey after 2017. The 2017 estimates are based on approximately 130,000 telephone interviews and carry state-level margins of error of  $\pm 3$ –6 percentage points. Gallup classifies a respondent as “very religious” if they report both that religion is important in their daily life *and* that they attend religious services weekly or almost weekly.

Note also that the DiD estimator in this paper relies on *within-state, cross-year* comparisons. Even if a high-religiosity state has a numerically higher normalized series than a low-religiosity state in levels, this does not affect the DiD estimate, since state fixed effects  $\mu_i$  absorb all time-invariant state-level differences. The identifying variation is entirely in *changes* across 2019 and 2020 within each state, not in levels across states.

### 0.1.3 An Empirical Test for Residual Ceiling Concerns

Although the ceiling effect is ruled out on design grounds, I provide an additional empirical check as a transparency exercise. The test examines whether the 2019 pre-treatment RSI baseline—measured over the days in  $D_{it} = 1$  from 2019—systematically differs between high- and low-religiosity states, and whether it is close to 100 in either group. If high-religiosity states had 2019 baselines near 100 under the joint normalization, that would indicate that 2019 search was driving the joint maximum, potentially leaving limited upward scope in 2020. Conversely, if 2019 baselines are well below 100 in both groups, the ceiling cannot be binding.

In Stata, this check can be implemented as:

```
* Summarize 2019 pre-treatment baseline by religiosity group
sum d_god if year == 2019 & post_treatment1 == 1 & high_relig == 1
sum d_god if year == 2019 & post_treatment1 == 1 & high_relig == 0

* Formal t-test of the difference in 2019 baselines
ttest d_god if year == 2019 & post_treatment1 == 1, by(high_relig)
```

Two findings from this exercise would jointly dismiss the ceiling concern: (i) neither group has a mean 2019 baseline close to 100, confirming that the joint maximum is dominated by 2020 observations; and (ii) even if high-religiosity states have a statistically higher 2019 baseline, the absolute level remains sufficiently below 100 that the ceiling cannot mechanically generate the observed 2.6-point differential treatment effect.

### 0.1.4 Economic Interpretation of the Heterogeneity Finding

Having established that the estimated heterogeneity coefficient  $\hat{\phi} = -2.61$  (s.e. 0.93,  $p = 0.006$ ) is not an artifact of measurement constraints, I now consider three economic mechanisms that could explain why high-religiosity states exhibited a *smaller* treatment effect than low-religiosity states, despite both groups showing large and significant increases in religious search.

**Extensive margin vs. intensive margin responses.** The most compelling interpretation centers on the distinction between *new* religious interest generated by the crisis and *existing* religious interest that was merely redirected online. In low-religiosity states, the share of the population with any prior engagement with religion is small. The shock of COVID-19—combining acute mortality salience, social isolation, and economic anxiety—may have drawn a non-trivial fraction of previously non-religious individuals to search for religious content for the first time. This is an *extensive margin* response: new participants entering the market for religious search. In high-religiosity states, by contrast, most residents already had well-established religious identities and routines. The lockdown disrupted the *form* of their religious engagement (shifting it from in-person worship to online) but did not generate fundamentally new demand. This is an *intensive margin* adjustment. Extensive margin responses are typically larger in magnitude—a person going from zero to positive religious search generates a larger measured increase than a regular churchgoer shifting from attending services to searching online for the same sermon. This mechanism implies that the differential treatment effect reflects genuine heterogeneity in the *type* of behavioral response, with Google Trends being a particularly sensitive instrument for detecting new entrants into religious activity.<sup>2</sup>

<sup>2</sup>This interpretation is consistent with the broader literature on religion and crises. ? model religious participation as yielding both consumption and afterlife utility. Under this framework, a large mortality salience shock would differentially affect individuals whose pre-shock probability of religious participation was near the margin, which by definition corresponds to residents of low-religiosity states.

**Institutional resilience and the digital pivot.** High-religiosity states are characterized by denser religious infrastructure: more churches per capita, stronger clergy networks, and more established congregational communities. When in-person services were prohibited, these communities were better positioned to execute a rapid *digital pivot*—transitioning to livestreamed services, online prayer groups, and church-specific apps and websites. If existing congregants followed their church’s digital presence directly (e.g., via a church website, a denomination’s app, or a WhatsApp group) rather than searching Google, this would suppress the measured search response in high-religiosity states without implying a smaller underlying religious engagement response. In low-religiosity states, individuals seeking religious content during the crisis would lack established institutional channels to turn to, making generic search engines the natural first port of call. Under this mechanism, the Google Trends measure *understates* the true religious engagement response in high-religiosity states relative to low-religiosity states, and the negative  $\hat{\phi}$  reflects a measurement artifact of the digital infrastructure gap rather than a genuine difference in behavioral response.

**Information-seeking vs. practice-seeking behavior.** A third, complementary mechanism concerns the *nature* of the searches driving the RSI. Search queries for terms like “God,” “Prayer,” or “Faith” can reflect two qualitatively distinct motivations: *information-seeking* (learning about religion, finding a local church, understanding religious concepts) and *practice-seeking* (finding a specific prayer, accessing a liturgical text, locating a streaming service for a known denomination). The former is more likely to be generated by individuals with low prior religious familiarity—residents of low-religiosity states who are encountering religious content for the first time during the crisis. The latter is more likely in high-religiosity states, where individuals have the doctrinal knowledge to search for specific content directly, generating fewer generic high-RSI queries of the type captured by broad topic searches. This mechanism implies that the measured RSI series is more sensitive to information-seeking behavior, and therefore disproportionately captures the response in low-religiosity states. It also suggests that more granular search data—e.g., subcategory-level topics such as “Christianity > Prayer” or “Islam > Quran”—might show a smaller differential between high- and low-religiosity states, since practice-seeking searches would be better captured by these narrower categories. This provides a testable implication that can be evaluated using the subcategory-level results reported in Section 4.

## References